SUBJECT : PROBLEM SOLVING AND TESTING USING JAVA

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Task 2 (Difficult): Real-Time Stream Analytics Engine

Problem Description

A cloud analytics platform receives millions of sensor readings. Each reading contains a sensor ID and temperature value. Using stream processing concepts, perform the following operations:

1. Filter temperatures greater than 50.
2. Group readings by sensor ID.
3. Compute average temperature per sensor.
4. Sort sensors based on average temperature in descending order.

Input Format

- First line contains integer N.
- Next N lines contain SensorID and Temperature.

Output Format

Display SensorID and average temperature sorted in descending order.

Constraints

- $1 \leq N \leq 10^5$
- $0 \leq \text{Temperature} \leq 100$

Sample Input

6
S1 60
S2 40
S1 80

S3 70
S2 90
S3 30

Sample Output

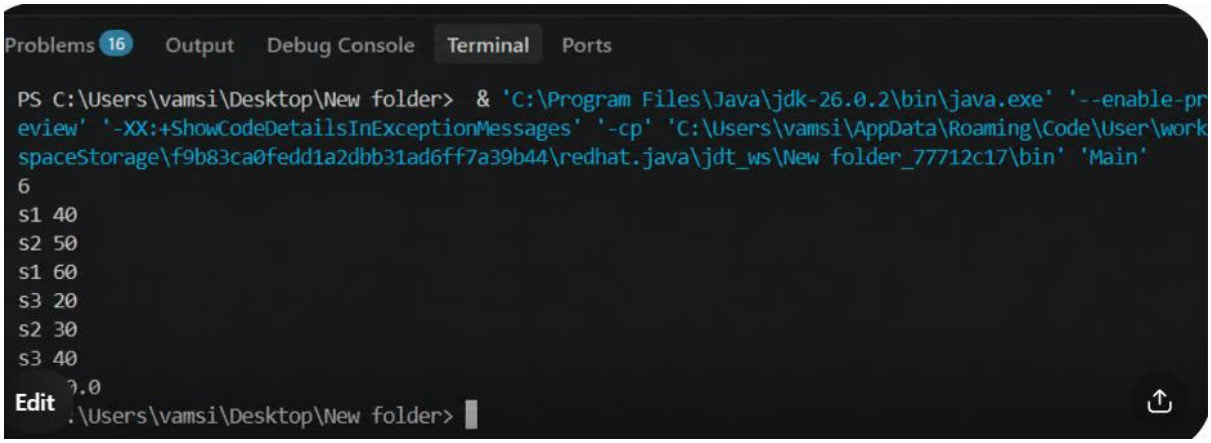
S1 70.0
S2 90.0
S3 70.0

PROGRAM :

```
import java.util.*;
import java.util.stream.*;

public class Main {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int n = sc.nextInt();

      Map<String, List<Double>> data = new HashMap<>();

      for (int i = 0; i < n; i++) {
         String sensor = sc.next();
         double temp = sc.nextDouble();

         // Filter temperature > 50
```

```
            if (temp > 50) {
                data.computeIfAbsent(sensor, k -> new ArrayList<>()).add(temp);
            }
        }

        // Calculate average and sort in descending order
        data.entrySet().stream()
            .map(e -> new AbstractMap.SimpleEntry<>(
                    e.getKey(),
                    e.getValue().stream()
                        .mapToDouble(Double::doubleValue)
                        .average()
                        .orElse(0)
            ))
            .sorted((a, b) -> Double.compare(b.getValue(), a.getValue()))
            .forEach(e -> System.out.println(
                    e.getKey() + " " + e.getValue()
            ));

        sc.close();
    }
}
```

OUTPUT :


Problems 16
Output
Debug Console
Terminal
Ports
PS C:\Users\vamsi\Desktop\New folder> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-pr
eview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\vamsi\AppData\Roaming\Code\User\work
spaceStorage\f9b83ca0fedd1a2dbb31ad6ff7a39b44\redhat.java\jdt_ws\New folder_77712c17\bin' 'Main'
6
s1 40
s2 50
s1 60
s3 20
s2 30
s3 40
Edit
.\Users\vamsi\Desktop\New folder>